

BERYL HOVIS-AFFLERBACH

ASTRONOMY PHD CANDIDATE

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EDUCATION

2023 - present	Northwestern University PhD candidate in Astronomy; advisor: Prof. Allison Strom Master of Science, Astronomy, June 2025	Evanston, IL
2018 - 2023	California Institute of Technology Bachelor of Science, Astrophysics, June 2023	Pasadena, CA

PUBLICATIONS

- The mass distribution of stars stripped in binaries: The effect of metallicity*
Hovis-Afflerbach, B., Göteborg, Y., Schootemeijer, A., Klencki, J., Strom, A. L., Ludwig, B. A., Drout, M. R., 2025, *A&A*, 697, A239
- Partially Erupted Prominence Material as a Diagnostic of Coronal Mass Ejection Trajectory*
Hovis-Afflerbach, B., Thompson, B. J., & Mason, E. I., 2023, *Space Weather*, 21, e2022SW003256
- Two New Methods for Counting and Tracking the Evolution of Polar Faculae*
Hovis-Afflerbach, B. & Pesnell, W. D., 2022, *Sol Phys*, 297, 48
- Identifying and Repairing Catastrophic Errors in Galaxy Properties Using Dimensionality Reduction*
Hovis-Afflerbach, B., Steinhardt, C. L., Masters, D., & Salvato, M. 2021, *ApJ*, 908, 148
- The BUFFALO HST Survey*
Steinhardt, C. L., et al., incl. **Hovis-Afflerbach, B.**, 2020, *ApJS*, 247, 1538

AWARDS & FUNDING

2023	NSF Graduate Research Fellow	NSF
2022	Vodopia-Hasson Poster Competition Award	Caltech
2022	George W. Housner Fund Recipient	Caltech
2021	Chambliss Undergraduate Poster Award, AAS 238	AAS
2021	Perpall Speaking Competition Semifinalist	Caltech
2021	Carnegie Observatories Summer Student Poster Award	Carnegie Observatories
2019	George W. Housner Fund Recipient	Caltech
2018	National Merit Scholar	NMSC
2018	NASA GSFC Intern Research Poster Session Finalist	NASA GSFC
2017	NASA GSFC Intern Research Poster Session Award	NASA GSFC

SELECTED ACADEMIC PRESENTATIONS

2026 / 05 Talk	Binary Products as Ionizing Sources: Testing & Comparing Models Across Cosmic Time The Metal-Poor Frontier: Understanding Low-Metallicity Stars and Galaxies Across Cosmic Time	STScI
2025 / 09 Talk	Finding He II: Testing Novel Models of Binary Populations Across Cosmic Time IAUS 402: Massive stars across redshifts in the era of JWST and large surveys	UNAM, Ensenada, B.C. México
2025 / 05 Poster	Testing Novel Models of Binary Populations Across Cosmic Time Inaugural Cosmic Frontier Center Conference	Cosmic Frontier Center, UT Austin
2024 / 05 Poster	Missing Massive Stripped Stars at Low Metallicity: Implications for High-Redshift Galaxies First Stars VII	Flatiron Institute
2024 / 02 Talk	Missing Massive Stripped Stars at Low Metallicity CIERA Theory Group meeting	Northwestern University
2023 / 01 Poster	A New Way to Model the Pre-Supernova Evolution of 8-11 $M_{\odot}$ Stars American Astronomical Society, AAS Meeting #241	AAS
2022 / 06 Poster	Testing Late Stellar Expansion with Populations of Stars Stripped in Binaries American Astronomical Society, AAS Meeting #240	AAS
2021 / 06 Poster	Two New Methods for Counting and Tracking the Evolution of Polar Faculae American Astronomical Society, AAS Meeting #238, won Chambliss Award	AAS

MENTORSHIP

2025 - present	CIERA mentorship network Mentor for freshman undergraduate student in astronomy.	Northwestern University
2022 - present	Mentor for undergraduate students in CASSI summer program As program alum, served as a mentor to 1-2 students each summer.	Carnegie Observatories
2024 - 2025	Physics & Astronomy graduate student peer mentoring Peer mentor for 2 first year graduate students.	Northwestern University

OUTREACH

2024 - present	Research Experiences in Astronomy at CIERA for High School Students (REACH) Developed and presented introductory lesson about stars including interactive component, served on panels, python coding tutor.	Northwestern University
2025 Volunteer	CIERA Reaches Out to Chicago Schools (CROCS) Visited local high schools and gave presentations and demonstrations on spectroscopy and gravitational waves.	Northwestern University
2024 - present Volunteer	Astronomy Conversations Held discussions with visitors to the planetarium and answered questions about astronomy, in the Adler Planetarium's Space Visualization Lab.	Adler Planetarium

2023 - present	Astronomy on Tap	Northwestern University
Volunteer, public talk	Helped organize events for Chicago chapter; gave public talk on binary stars.	
2020 & 2021	Upward Bound College Panel	Carnegie Observatories
Panelist	Answered questions from incoming first-generation college students about college, admissions, and astronomy.	
2021	International Women's Day Panel	Girl Up Club at Highland Park High School
Panelist	Talked about careers in astronomy and answered questions about research, mentorship, and succeeding as a woman in STEM.	

LEADERSHIP & SERVICE

2026 - present	Referee, The Astrophysical Journal (ApJ)	AAS
2025 - present	CIERA Stars Meeting	Northwestern University
	Started and organized biweekly meeting for those interested in stars from different perspectives to share research, discuss recent papers and results, and find opportunities for collaboration.	
2025 - present	President, Physics & Astronomy Graduate Student Council	Northwestern University
2025 - 2026	Science Editor, CIERA 15th Anniversary Coffee Table Book	Northwestern University
2024 - 2025	Secretary, Physics & Astronomy Graduate Student Council	Northwestern University
2019 - 2022	Equity & Title IX Advocate	Caltech
	Provided support for peers in cases relating to Equity & Title IX violations. Organized meetings to welcome new students and share Title IX resources.	
2019 - 2022	Board of Control Representative	Caltech
	Heard cases regarding alleged academic Honor Code violations, >100 hours.	
2021	CASSI DEI Programming	Carnegie Observatories
	Helped lead DEI discussions for Carnegie Observatories summer program.	

TEACHING EXPERIENCE

2025 / Spring	ASTRON 220-1 — Intro to Astrophysics I	Northwestern University
	Teaching assistant for course on the origin and evolution of star/planet systems, focusing on underlying physical processes and observational techniques.	
2025 / Winter	PHYSICS 135-3 — General Physics	Northwestern University
	Teaching assistant for introductory waves & optics.	
2024 / Fall	PHYSICS 135-2 — General Physics	Northwestern University
	Teaching assistant for introductory electricity & magnetism.	

SOFTWARE & LANGUAGES

Python • MESA • Cloudy • Pypelt • Linux • IDL • HTML • CSS